On the geometry of Coulomb branches

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Slides:
https://bwebste.github.io/Glasgow-2026.pdf

Fix a reductive group G and representation $\mathbf{N}$. The **Coulomb branch** $\mathcal{M}(G,\mathbf{N})$ is a symplectic variety constructed via quantum field theory.

Two main questions

1. How can we describe Coulomb branches geometrically in a way that is *uniform* across the rational, K-theoretic, and elliptic settings?
2. $\mathcal{M}(G,\mathbf{N})$ has symplectic singularities, so it has finitely many **symplectic leaves**. What is their classification, and what are their transverse slices?

Based on draft paper
`https://bwebste.github.io/GoC.pdf`

An important 3d theory: given G a connected reductive group and $\mathbf{E}$ a symplectic representation of G, we have a theory $\mathcal{T}(G, \mathbf{E})$ given by gauging the 3d σ-model with target $\mathbf{E}$. Most often $\mathbf{E} = T^*\mathbf{N}$ for a representation $\mathbf{N}$, but I don't want to assume this right away.

We have two natural topological twists Q_A, Q_B. We call theories **3-d mirror dual** if they are equivalent in a way that swaps $Q_A \leftrightarrow Q_B$.

Since theories are 3d, their local operators are commutative rings equipped with Poisson brackets. We think of these as functions on algebraic varieties called the **Higgs** and **Coulomb** branches $\mathfrak{M}_H$ and $\mathfrak{M}_C$.

The moduli of vacua $\mathfrak{M}_H(G, \mathbf{E}) = \operatorname{Spec} Z_B(S^2)$ for the B-twist (the **Higgs branch**) was studied by Hitchin, Karlhede, Lindström and Roček and interpreted as a hyperkähler quotient $\mathbf{E} /\!\!/\!\!/\!\!/ G$.

The moduli of vacua for the A-twist (the **Coulomb branch**) proved much more mysterious. Denote this $\mathfrak{M}_C(G, \mathbf{N}) = \operatorname{Spec} Z_A(S^2)$ (I'm assuming $\mathbf{E} = T^*\mathbf{N}$).

The space $\mathfrak{M}_C(G, \mathbf{N})$ is birational to $T^*T^\vee/W$, but with a modification of the multiplication of functions on this space.

Work of Cremonesi–Hanany–Zaffaroni, Bullimore–Dimofte–Gaiotto, and Braverman–Finkelberg–Nakajima gradually brought this picture into clearer view:

- $\mathbb{C}[\mathfrak{M}_C(G,\mathbf{N})]$ is a subalgebra of the fraction field $\mathbb{C}(T^*T^\vee)^W$,
- obtained by replacing e^λ, a character on $T^\vee$, by multiplying it by certain rational functions (roughly with denominator depending on G and numerator depending on $\mathbf{N}$).

We can think of this as an affine version of Springer theory:

Taylor series $\mathsf{C} = \mathbb{C}[[t]]$	$\mathsf{G} = G[[t]]$	$\mathsf{N} = \mathbf{N}[[t]]$
Laurent series $\mathcal{C} = \mathbb{C}((t))$	$\mathcal{G} = G((t))$	$\mathcal{N} = \mathbf{N}((t))$

Relevant spaces:

$$\mathsf{Y} = \mathsf{N}/\mathsf{G} = \mathrm{Map}(D = \operatorname{Spec}\mathsf{C} \to \mathbf{N}/G)$$

$$\mathcal{Y} = \mathcal{N}/\mathcal{G} = \mathrm{Map}(D^* = \operatorname{Spec}\mathcal{C} \to \mathbf{N}/G)$$

These can be interpreted as spaces of principal G bundles with a section of the associated $\mathbf{N}$-bundle on D and D^*.

Thus, the fiber product $\mathsf{Y} \times_{\mathcal{Y}} \mathsf{Y}$ is the space of such bundles on the “raviolo” gluing two copies of D along D^*.

Previous experience tells us it would be fun to consider

$$A = H_*^{BM}(\mathsf{Y} \times_{\mathcal{Y}} \mathsf{Y}).$$

Using factorization arguments, we can see that A is a commutative $\mathbb{C}$-algebra of finite type.

Definition

The **Coulomb branch** *is the spectrum* $\mathcal{M} = \operatorname{Spec} A$.

This geometric definition has advantages, but it's very scary when you open the paper.

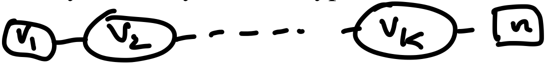
Most important case: let Γ be a quiver, and $\mathbf{v}\colon I = \mathcal{V}(\Gamma) \to \mathbb{Z}_{\geq 0}$ a dimension vector. Let

$$N_{\mathbf{v}} = \bigoplus_{i \to j} \mathrm{Hom}(\mathbb{C}^{v_i}, \mathbb{C}^{v_j}) \qquad G_{\mathbf{v}} = \prod_i GL(\mathbb{C}^{v_i})$$

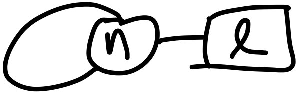
These recover interesting varieties, always symplectic:

- my favorite: nilcone of $\mathfrak{gl}_n$

- more generally, Slodowy slices in type A



- symmetric power $\mathrm{Sym}^n(\mathbb{C}^2/\mathbb{Z}_\ell)$



The BFN construction uses Borel–Moore homology. The same construction works uniformly for three choices of a 1-dimensional algebraic group B:

B	Cohomology theory	Base $\mathcal{B}_G$
$\mathbb{G}_a = \operatorname{Spec} \mathbb{C}[x]$	Rational (BFN)	$\mathfrak{t}/W$
$\mathbb{G}_m = \operatorname{Spec} \mathbb{C}[z^{\pm 1}]$	K-theoretic	T/W
E (elliptic curve)	Elliptic	$E \otimes_{\mathbb{Z}} X_*(T)/W$

In each case, $\mathcal{M}(G, \mathbf{N})$ is a variety over the **base** $\mathcal{B}_G$. For each weight φ of $\mathbf{N}$ (or root α of G), there is an **Euler class** $\operatorname{Eu}(\varphi) \in \mathcal{O}(\mathcal{B}_T)$ (linear / multiplicative / theta section), with divisor of zeros D_φ.

In the case $G = T$, $\mathcal{M}(T, \mathbf{N})$ is the spectrum of the subalgebra of $\mathcal{O}_{\mathcal{B}_T} \otimes_{\Bbbk} \Bbbk[T^\vee]$ generated by $\mathcal{O}_{\mathcal{B}_T}$ and $e^\lambda \prod_\varphi \mathrm{Eu}(\varphi)^{\max(\lambda(\varphi),0)}$ for $\lambda \in X^*(T)$ a cocharacter of T.

We can see this geometrically as adding extra divisors to $\mathcal{B}_T \times T^\vee$ which force these factors of Euler classes to appear.

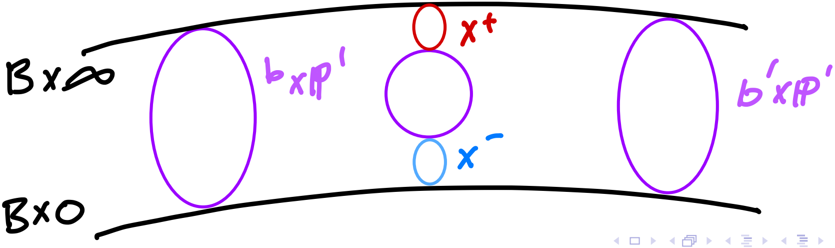
For example: $G = \mathbb{G}_m$. The weights of $\mathbf{N}$ are $n_1, \dots, n_k \in \mathbb{Z}$. Let

$$\mathsf{Eu}^{\pm} = \prod_{\pm n_i > 0} \mathrm{Eu}(n_i).$$

In $B \times \mathbb{P}^1_z$, consider the subshemes $X^{\pm} = Z(\mathsf{Eu}^{\pm}, z^{\mp 1})$.

Theorem

$\mathcal{M}(\mathbb{G}_m, \mathbf{N})$ is the affinization of the blowup of $B \times \mathbb{P}^1_z$ at $X^+ \cup X^-$ with strict transforms of $B \times \{0\}$ and $B \times \{\infty\}$ removed.



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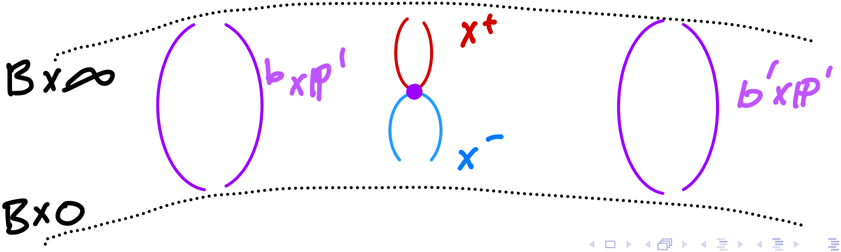


General case: need to do a similar blowup for each of the lines $\ell_1, \ldots, \ell_k$ spanned by weights of $\mathbf{N}$.

Teleman gluing

If the weights of $\mathbf{N}$ are generic enough, then all the new divisors can be written as the images of fibers over points in $\mathcal{B}_T$ under a birational map, so $\mathcal{M}(T, \mathbf{N})$ is the affinization of $\mathcal{B}_T \times T^\vee$ glued to itself along this birational map.

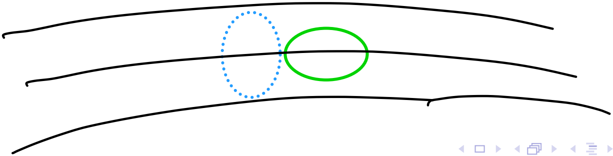
Unifies rational, K-theoretic, and elliptic cases. Local-to-global: K-theoretic and elliptic Coulomb branches look locally like the rational one (base $\cong$ tangent space locally).

Passing from T to G requires a further birational modification: if no root lies in ℓ_i, then we blow up the subscheme in $\mathcal{B}_T \times T^\vee$ given by $Z(\mathrm{Eu}(\alpha), e^\alpha - 1)$ for each root α, and remove the strict transform of $Z(\mathrm{Eu}(\alpha))$.

Theorem

The Coulomb branch $\mathcal{M}(G, \mathbf{N})$ is the affinization of this birational modification of $\mathcal{B}_T \times T^\vee$ modulo W.

If roots do lie in ℓ_i, then fixable, but need to be more careful.



Put differently: $\mathcal{M}(G, \mathbf{N})$ is a normal variety, so we can describe it by giving Cartier divisors. These are only different from $(\mathcal{B}_T \times T^\vee)/W$ over the divisors where Euler classes of roots or weights vanish, and the modification is obtained from those divisors in a uniform geometric way.

- Over a weight divisor where no root vanishes, we have the two divisors coming from the blowups as in the abelian case.
- Over a root divisor, we usually only have one divisor, since the reflection combines them (though there are a couple of exceptions where there are two).

Important consequence: removing a divisor has the same effect as removing the corresponding weight from $\mathbf{N}$ or root from G.

Given $\lambda \in \mathfrak{t}$, there is a commuting subgroup $G_\lambda \subset G$ with Weyl group W_λ and fixed representation $\mathbf{N}^\lambda \subset \mathbf{N}$.

Let $U \subset \mathcal{B}_G$ be the complement of the divisors corresponding to roots or weights which don't vanish at λ.

Theorem

The varieties $\mathcal{M}(G,\mathbf{N})$ and $\mathcal{M}(G_\lambda,\mathbf{N}^\lambda)$ over U/W_λ are isomorphic.

$$\begin{tikzcd}
\mathcal{M}(G,\mathbf{N}) \arrow[d] & \mathcal{M}^\circ(G_\lambda,\mathbf{N}_\lambda) \arrow[l] \arrow[r, hook] \arrow[d] & \mathcal{M}(G_\lambda,\mathbf{N}_\lambda) \arrow[d] \\
\mathcal{B}_G & U/W_\lambda \arrow[l] \arrow[r, hook] & \mathcal{B}_{G_\lambda}
\end{tikzcd}$$

Bielawski–Foscolo proposed describing Coulomb branches via **transverse W-Hilbert schemes**.

Closure of free W-orbits in the moduli of subschemes of $\mathcal{M}(T, \mathbf{N})$ that push forward to subscheme of $\mathcal{B}_T$.

Theorem

The functions on $\mathrm{Hilb}^W_\pi(X)$ *are the subring of* $\Bbbk(X)^W$ *generated by* $(\frac{1}{\Delta}\Bbbk[X])^W$.

For $G = PGL(2)$ or a quiver gauge theory: exactly $\mathcal{M}(G, \mathbf{N})$.

The issue for general G: there "aren't enough" sign-invariant functions on $\mathcal{M}(T,\mathbf{N})$ to generate the functions on $\mathcal{M}(G,\mathbf{N})$.

Corrected theorem

Replace $\mathcal{M}(T,\mathbf{N})$ by the open subset $\mathcal{M}^\bullet(T,\mathbf{N})$ obtained by removing the locus over intersections of bad divisors. Then:

$$\mathcal{M}^\bullet(G,\mathbf{N}) \cong \mathrm{Hilb}_\pi^W(\mathcal{M}^\bullet(T,\mathbf{N}))$$

and $\mathcal{M}(G,\mathbf{N})$ is the affinization of this variety.

A **symplectic singularity** has a canonical stratification into finitely many **symplectic leaves**: locally closed smooth symplectic subvarieties (Kaledin). These are the orbits of the Hamiltonian vector fields.

Fundamental questions for $\mathcal{M}(G, \mathbf{N})$:

1. Index the leaves (how many are there? how to describe them?).
2. Identify the **transverse slice** to each leaf (another symplectic singularity).

In the abelian case $G = T$, the answer is well-known from Higgs branch description:

- Leaves are in bijection with **co-loop-free flats** of $\mathfrak{t}$.
  - A **flat** F is an intersection of hyperplanes defined by weights of $\mathbf{N}$; this is precisely the stabilizer of a point in $\mathbf{N}$.
  - A **co-loop** of F is a weight φ such that there exists a cocharacter λ of T with $\lambda(\varphi) = 1$ and $\lambda(\psi) = 0$ for all other weights ψ vanishing on F.
- If λ is generic in this stabilizer T_0, then the slice to this leaf is $\mathcal{M}(T/T_0, \mathbf{N}^\lambda)$.

If a flat has a co-loop, it still gives a subspace, but not a symplectic leaf: consider $\mathcal{M}(\mathbb{G}_m, \mathbb{C}) \cong \mathbb{C}^2$.

Quite suggestive, but it simplifies the general case quite a bit.

The distinction between good, bad and ugly theories will be important for us:

- **good**: the scaling action contracts to a unique fixed point, which is a 0-dimensional symplectic leaf. Example: $\mathcal{M}(\mathbb{G}_m, \mathbb{C}^2) \cong \mathbb{C}^2/\mathbb{Z}_2$, abelian theories with finite kernel, no co-loops.
- **ugly**: the scaling action contracts to a fixed point, but this fixed point lies in a positive-dimensional leaf (in fact a symplectic vector space). Example: $\mathcal{M}(\mathbb{G}_m, \mathbb{C}) \cong \mathbb{C}^2$, abelian theories with finite kernel and co-loops.
- **bad**: the scaling action doesn't contract to a single fixed point. Example: $\mathcal{M}(\mathbb{G}_m, 0) \cong \mathbb{C} \times \mathbb{C}^\times$, abelian theories with infinite kernel.

Abelian case: a leaf appears exactly when $(T/T_0, \mathbf{N}^\lambda)$ is good.

There is a natural **integrable system**:

$$\varrho\colon \mathcal{M}(G,\mathbf{N}) \longrightarrow \mathcal{B}_G,$$

induced by the map to the pure Coulomb branch $\mathcal{M}(G)$, which is a group scheme over $\mathcal{B}_G$.

Over the open dense set of $\mathcal{B}_G$ where no weights or roots vanish, the fiber of ϱ is a torsor for the dual torus $T^\vee$.

Can break the classification of leaves into two steps:

1. Classify the possible images of leaves in $\mathcal{B}_G$.
2. For each image, classify the leaves lying over it.

Definition

A **flat** *of the generalized root hyperplane arrangement in $\mathfrak{t}$ is an intersection of a subset of the hyperplanes H_φ defined by weights of* $\mathbf{N}$*; these are exactly the stabilizers of points in* $\mathbf{N}$.

For a flat $U \subset \mathfrak{t}$ with generic point λ:

- $G_\lambda = C_G(\lambda)$ is a **Levi subgroup** of G
- $\mathbf{N}^\lambda = \mathbf{N}^\lambda$ is the fixed subspace of λ (a G_λ-representation)
- $G_\lambda^0 = \ker^0(G_\lambda \to GL(\mathbf{N}^\lambda))$; $\quad G_\lambda^1 = G_\lambda / G_\lambda^0$ almost faithful on $\mathbf{N}^\lambda$
- $\mathfrak{t}_\lambda^0 =$ Lie algebra of maximal torus of G_λ^0 (of dimension k)

Slice theorem (local structure)

The fiber of ϱ over λ is étale locally symplectomorphic to $\mathcal{M}(G_\lambda^1, \mathbf{N}^\lambda) \times \mathbb{A}^{2k}$.

We have to be a little more careful about global structure, though.

- There might be a Galois group $\bar{\Gamma}_\lambda$ that appears in the comparison of $\mathcal{M}(G, \mathbf{N})$ and $\mathcal{M}(G_\lambda, \mathbf{N}^\lambda)$ over U.
- There is an isogeny $G_\lambda \to G^1_\lambda \times \bar{G}^0_\lambda$ for $\bar{G}^0_\lambda$ a finite quotient; let Z_λ be the kernel of this map.

Lemma

The product $\mathcal{M}(G^1_\lambda \times \bar{G}^0_\lambda, \mathbf{N}^\lambda) \cong \mathcal{M}(G^1_\lambda, \mathbf{N}^\lambda) \times \mathcal{M}(\bar{G}^0_\lambda, 0)$ admits a free action of $Z^\vee_\lambda$, and $\mathcal{M}(G_\lambda, \mathbf{N}^\lambda)$ is the quotient by this action.

The symplectic leaves of $\mathcal{M}(G_\lambda, \mathbf{N}^\lambda)$ are in bijection with $Z^\vee_\lambda$-orbits on the symplectic leaves of $\mathcal{M}(G^1_\lambda, \mathbf{N}^\lambda)$, and the transverse slice to a leaf in $\mathcal{M}(G_\lambda, \mathbf{N}^\lambda)$ is the same as the transverse slice to any leaf in the corresponding orbit in $\mathcal{M}(G^1_\lambda, \mathbf{N}^\lambda)$.

Theorem (W.)

1. *The image of any symplectic leaf of $\mathcal{M}(G, \mathbf{N})$ under ϱ is the image in $\mathfrak{t}/W$ of a flat U.*
2. *The leaves with image U are in bijection with $Z_\lambda^\vee \rtimes \bar{\Gamma}_\lambda$-orbits on 0-dimensional leaves of $\mathcal{M}(G_\lambda^1, \mathbf{N}^\lambda)$.*
3. *If U is* ***good*** *(i.e., $\mathcal{M}(G_\lambda^1, \mathbf{N}^\lambda)$ is good), there is a* ***unique*** *leaf with image U, and its* ***transverse slice*** *is isomorphic to $\mathcal{M}(G_\lambda^1, \mathbf{N}^\lambda)$.*
4. *If $\mathcal{M}(G_\lambda^1, \mathbf{N}^\lambda)$ has no 0-dimensional leaves, there is* ***no*** *leaf with image U. Important case: $\mathcal{M}(G_\lambda^1, \mathbf{N}^\lambda)$ is ugly.*

Lemma

A leaf L with image U lies in the closure of L' with image U' if and only if:

1. $U \subset U'$, *and*
2. *the corresponding closure relation holds inside* $\mathcal{M}(G^1_\lambda, \mathbf{N}^\lambda)$.

In particular, if both U and U' are good, then $L \subset \overline{L'}$ if and only if $U \subset U'$.

This gives a complete description of the **closure poset** on good leaves: reverse inclusion on good flats.

Consider the case $G = \mathrm{GL}_n$, $\mathbf{N} = \mathfrak{gl}_n$.

1. $\mathcal{M}(G, \mathbf{N}) \cong \mathrm{Sym}^n(\mathbb{C} \times \mathbb{C}^\times)$.
2. Flats: indexed by partitions $\lambda \vdash n$ (groups of equal coordinates).
3. For $\lambda = (n_1, \ldots, n_k)$:

$$G^1_\lambda = \prod PGL_{n_i}, \quad \mathbf{N}^\lambda = \bigoplus \mathfrak{gl}_{n_i}.$$

4. Residual theory $\mathcal{M}(G^1_\lambda, \mathbf{N}^\lambda)$ has $\prod n_i$ zero-dimensional leaves, but $Z^\vee$ acts simply transitively $\Rightarrow$ **one leaf per partition**.

These are special cases of quiver gauge theories.

$$G_{\mathbf{v}} = \left(\prod_i \mathrm{GL}_{v_i}\right)/\mathbb{C}^\times \qquad \mathbf{N}_{\mathbf{v}} = \bigoplus_{i\to j} \mathrm{Hom}(\mathbb{C}^{v_i}, \mathbb{C}^{v_j})$$

A cocharacter for a quiver gauge theory is a grading of the corresponding quiver representation, so the flat is described by the dimension vectors of the graded pieces.

"Fission and decay"

Writing $\mathbf{v} = \sum \mathbf{v}^{(i)}$ as a sum of dimension vectors:

1. "Decay": If $\mathbf{v}^{(i)}$ is a unit vector, trivial contribution to the slice.
2. "Fission": If more than one of the summands $\mathbf{v}^{(i)}$ is not a unit vector, then the slice is the product of contributions from the nontrivial $\mathbf{v}^{(i)}$.

Now modify to $G = \mathrm{GL}_n$, $\mathbf{N} = \mathbb{C}^n \oplus \mathfrak{gl}_n$.

1. $\mathcal{M}(G, \mathbf{N}) \cong \mathrm{Sym}^n(\mathbb{C}^2)$.
2. Flats: indexed by an integer $n_0 \leq n$ (number of coordinates equal to 0) and by partitions $\lambda \vdash (n - n_0)$.
3. Slice described by

$$G_\lambda^1 = GL_{n_0} \times \prod PGL_{\lambda_i}, \quad \mathbf{N}^\lambda = \mathbb{C}^{n_0} \oplus \mathfrak{gl}_{n_0} \oplus \bigoplus \mathfrak{gl}_{\lambda_i}.$$

   $(\mathrm{GL}_{n_0}, \mathbb{C}^{n_0} \oplus \mathfrak{gl}_{n_0})$ is **ugly** for all $n_0 > 0$, so $\mathcal{M}(G_\lambda^1, \mathbf{N}^\lambda)$ has no 0-dimensional leaves in this case.
4. For $n_0 = 0$, same flats as above give the same Hasse diagram.

One more: $G = \mathrm{GL}_n$, $\mathbf{N} = (\mathbb{C}^n)^{\oplus \ell} \oplus \mathfrak{gl}_n$ ($\ell > 1$).

1. $\mathcal{M}(G, \mathbf{N}) \cong \mathrm{Sym}^n(\mathbb{C}^2/\mathbb{Z}_\ell)$.
2. Flats indexed by pairs (n_0, λ) with $n_0 \leq n$, $\lambda \vdash (n - n_0)$.
3. Slice described by

$$G^1_\lambda = GL_{n_0} \times \prod PGL_{\lambda_i}, \quad \mathbf{N}^\lambda = (\mathbb{C}^{n_0})^{\oplus \ell} \oplus \mathfrak{gl}_{n_0} \oplus \bigoplus \mathfrak{gl}_{\lambda_i}.$$

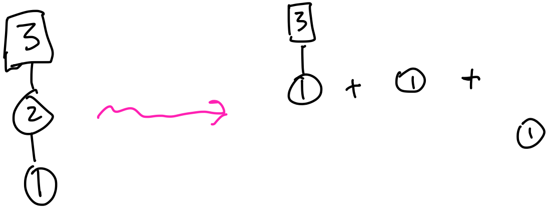
4. For $n_0 > 0$, $(\mathrm{GL}_{n_0}, (\mathbb{C}^{n_0})^{\oplus \ell} \oplus \mathfrak{gl}_{n_0})$ is **good**. So there is a leaf for each pair (n_0, λ), given by n_0 points at the origin and $n - n_0$ points in $\mathbb{C}^2/\mathbb{Z}_\ell$ with multiplicities given by λ.

Examples

Note that the framing is expressed by adding a new vertex, so it all goes with one summand.

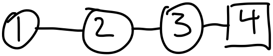
For a framed Dynkin quiver, the only contributions come from dimension vectors of dominant weights, with unit vectors "decayed off."

This matches the results of Muthiah and Weekes.



For example: cotangent bundle of a type A flag variety:

- Theory is $G = \mathrm{GL}_1 \times \cdots \times \mathrm{GL}_{n-1}$, $\mathbf{N} = \bigoplus_{i=1}^{n-1} \mathrm{Hom}(\mathbb{C}^i, \mathbb{C}^{i+1})$.

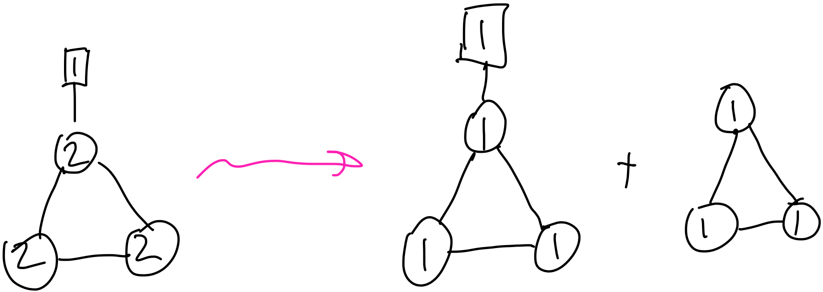


- The good residual theories correspond to dimension vectors v_i with $v_1 \leq v_2 - v_1 \leq \cdots \leq v_{n-1} - v_{n-2} \leq n - v_{n-1}$. These give a partition $\lambda_i = v_{n-i+1} - v_{n-i}$.

$\lambda = (2, 1, 1)$

The corresponding leaf has Jordan type λ'; the slice $\mathcal{M}(PGL_{\mathbf{v}}, \mathbf{N}_{\mathbf{v}})$ is the Slodowy slice to the nilpotent orbit of type λ'.

For an affine type A quiver, we can also “fission” off multiples of the imaginary root δ, contributing the centered Uhlenbeck space $\mathcal{M}(PGL_{k\delta}, \mathbf{N}_{k\delta})$ to the slice, matching Nakajima–Takayama.



Let $\mathrm{Leaf}(\mathcal{M}_{\mathsf{Coulomb}})$ and $\mathrm{Leaf}(\mathcal{M}_{\mathsf{Higgs}})$ denote the leaf sets of the Coulomb and Higgs branches.

Definition

Define $\mathbf{CH} \subset \mathrm{Leaf}(\mathcal{M}_{\mathsf{Coulomb}}) \times \mathrm{Leaf}(\mathcal{M}_{\mathsf{Higgs}})$ *by* $(L, L') \in \mathbf{CH}$ *if, for a generic point* $[x] \in L'$*, the image of* L *under* ϱ *equals the flat* $\mathfrak{t}_x$ *(Lie algebra of the torus of the stabilizer* G_x*).*

Physics expects an **order-reversing bijection** between leaves, but several obstructions arise:

1. Multiple Coulomb leaves over the same flat.
2. A flat may not arise as $\mathfrak{t}_x$ for any Higgs leaf.
3. Multiple Higgs leaves with the same stabilizer torus.

Example: $G = \mathrm{GL}_n$, $\mathbf{N} = \mathbb{C}^n \oplus \mathfrak{gl}_n$. Every Higgs leaf has $\mathfrak{t}_x = \mathfrak{t}$, so only the open Coulomb leaf appears.

Conjecture

There is a natural notion of ***special*** *leaf on each side such that* **CH** *restricts to a bijection between the special leaves of* $\mathcal{M}_{\mathsf{Coulomb}}$ *and* $\mathcal{M}_{\mathsf{Higgs}}$.
The special Coulomb leaves are exactly those whose image under ϱ *is a* ***good*** *flat.*

Evidence:

- **Toric case** ($G = T$): all leaves are special; bijection is obvious.
- **Dynkin quivers**: all leaves are special; bijection is obvious.
- $G = \mathrm{GL}_n$, $\mathbf{N} = (\mathbb{C}^n)^{\oplus \ell} \oplus \mathfrak{gl}_n$: special leaves are those with $\lambda = (1, \dots, 1)$. Order-reversing bijection with special Higgs leaves, but not all leaves.

1. **Zero-dimensional leaves in bad/ugly theories.** When does a bad flat admit a 0-d leaf? Often none exist, but sometimes they do. For $G = PGL_n$, $\mathbf{N} = \mathfrak{pgl}_n$ shows that any number of zero-dimensional leaves is possible.
2. **Elliptic Coulomb branches.** No consensus definition exists yet. Our construction gives a candidate; comparison with BFN is pending more development in the elliptic literature.
3. **Non-symmetric quivers (Nakajima–Weekes).** Construction of slices should be exactly the same, but I have not looked carefully at any examples.